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Glomerospores: a new denomination for the spores of Glomeromycota, a group molecularly distinct from the Zygomycota

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Abstract—The arbuscular mycorrhizal fungi (AMF) are grouped into the Glomeromycota phylum. However, the names used for the spores formed by these fungi are not clear, being based on morphological characters, some of which can be shared by fungi in separate phyla (Glomeromycota and Zygomycota – *Endogonales*), due to adaptative convergence. Considering also that all other phyla of fungi have their own characteristic spore types, with unique names, we propose a denomination to be used only for the spores of Glomeromycota.

Key Words—Glomeromycetes, mycorrhiza, taxonomy, terminology

Taxonomy

The terminology used for the spores of arbuscular mycorrhizal fungi (AMF) has changed according to each author's view, and different names such as chlamydospores, conidia, sporophores, azygospores or even simply spores have been used to describe these propagative structures (Gerdemann & Trappe 1974, Gibson 1985, Walker & Sanders 1986, Walker 1987, Morton 1988, Schenck & Pérez 1990, Tommerup & Sivasithamparam 1990).

Nowadays the AMF, earlier included in *Glomales*, an order of Zygomycetes (Morton & Benny 1990), are separated from the Zygomycota based on molecular studies. Schüssler et al. (2001) proposed the phylum Glomeromycota and indicated that these fungi are more closely related to Ascomycota and Basidiomycota than to Zygomycetes. This proposal has been accepted (Tehler et al. 2003), although for some it is premature to recognize the glomalean fungi as a phylum (Tanabe et al. 2005). Anyway, each fungal phylum has an appropriate name for its spores: ascospore for Ascomycota, basidiospore for Basidiomycota, zygospore for Zygomycota (Gerdemann & Trappe 1974, Alexopoulos et al. 1996). All of these are formed by sexual reproduction, while in Glomeromycota there is no evidence of a sexual process. On the other hand, asexual spores have different denominations, based on formation and similarity with spores of other groups.

Considering that in Glomeromycota the reproductive structures are apparently of asexual origin, there is no agreement among authors in regard to the nature of these spores (Morton 1988). The term chlamydospore, for example, was designated for the resistant spores of other species (Gerdemann & Trappe 1974). In *Acaulospora*, *Entrophospora* and *Gigaspora* the spores have been named azygospores, due to the similarity of the vesicle in the first two genera and the suspensor in the third, with the azygospores formed parthenogenetically in some Zygomycetes, a group in which the AMF were included.

This terminology is confused because within a phyletic group (Morton & Benny 1990, Schüssler et al. 2001), with reproduction exclusively asexual, the spores present diverse nature. This is not uncommon among other groups of asexual fungi such as the “Deuteromycetes”, that form chlamydospores, blastospores, arthrospores and conidia (Alexopoulos et al. 1996).

Endogone Link contains fungi morphologically similar to the Glomeromycota, but its spores (zygospores), produced in sporocarps, have sexual origin (Alexopoulos et al. 1996), while azygospores are rarely observed (Berch & Fortin 1983). During zygospore formation, two walls are distinguished: an outer wall, from the zygosporangium, and an inner wall, from the zygosporangium (O'Donnell et al. 1978a, b). In the AMF, the spore presents a single unique wall (Maia & Kimbrough 1993) formed by many layers of different appearance (Walker 1983).

Germination of spores in *Endogonales* is similar to that of Glomeromycetes (*Acaulospora*, *Entrophospora*, *Scutellospora* and *Pacispora*), with the germ tube emerging either from a germinal shield, directly through the spore wall or from the hyphal insertion point (O'Donnell 1978a, b).

The germinal shield of AMF is highly specialized, being helicoid (*Scutellospora projecturata* Kramad. & C. Walker) or rich in foldings (*Entrophospora colombiana* Spain & N.C. Schenck, *Acaulospora tuberculata* Janos & Trappe, *A. rehmsii* Sieverd. & S. Toro, *S. gilmorei* (Trappe & Gerd.) C. Walker & F.E. Sanders, *S. scutata* C. Walker & Dieder., *S. cerradensis* Spain & J. Miranda). Spores of *Endogone pisiformis* Link have a germinal shield less elaborated than that of the AMF, formed between the endospore wall (zygosporangium) and the exospore wall (zygosporangium) and although each spore can have up to three germinal shields, only one will originate the germinal tube.

On the other hand, spores of AMF produce a germinal shield in various places in the outer or innermost layers of the so-called “germinal wall”. Berch & Fortin (1983) mentioned that spores of AMF and *Endogone* present such characteristics by adaptive convergence, since in both cases the thick spore wall provides higher resistance to environmental stresses such as predation and action of soil microorganisms. Molecular data (Schüssler et al. 2001) seem to confirm this hypothesis of adaptive convergence, since the AMF form a homogeneous group separated from the Zygomycetes.

Chlamydospores are resistant spores, with a thick wall (Kirk et al. 2001). Based on molecular data (Schüssler et al. 2001) that show a monophyletic group, and also on the morphological diversity of the complex sub cellular structure of the AMF spore wall, unique in the Regnum Fungi, we propose the name glomerospores for the spores of Glomeromycota. This would distinguish these spores from those of other different taxa.

Glomerospores – Asexual reproductive structures formed at the end of a subtending hypha, or a bulbous sporogenous cell, or from a sporiferous saccule, in this case with lateral or intercalary position, with or without a germinative component, presenting or not a germinal shield or orb, typically formed by fungi that form an arbuscular mutualistic symbiosis with most terrestrial plants.

Etymology: “Glomero” = referring to the Glomeromycota phylum + “spore” = reproductive structure.

Acknowledgements

Thanks are due to Dr. R. Hanlin and J.W. Kimbrough for reviewing the manuscript and making helpful suggestions. The authors are in debt to Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) for providing PhD scholarship and research grant, respectively.

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